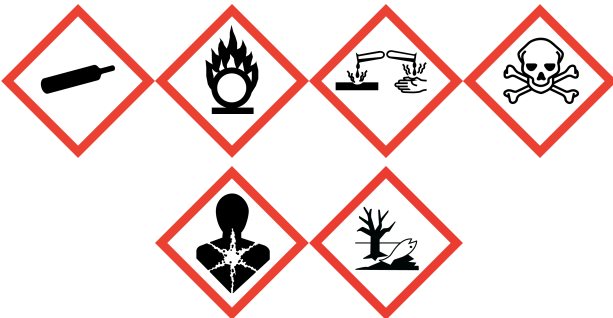


CHLORINE**ICSC: 0126 (March 2009)****CAS #: 7782-50-5****UN #: 1017****EC Number: 231-959-5**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible but enhances combustion of other substances. Many reactions may cause fire or explosion. Risk of fire and explosion. See Chemical Dangers.	NO contact with incompatible materials: See Chemical Dangers	In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep cylinder cool by spraying with water. NO direct contact with water.

AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Sore throat. Shortness of breath. Wheezing. Laboured breathing. Symptoms may be delayed. See Notes.	Use breathing protection, closed system or ventilation.	Fresh air, rest. Half-upright position. Refer immediately for medical attention. Artificial respiration may be needed. See Notes.
Skin	ON CONTACT WITH LIQUID: FROSTBITE. Redness. Burning sensation. Pain. Skin burns.	Cold-insulating gloves. Protective clothing.	First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Refer immediately for medical attention.
Eyes	Watering of the eyes. Redness. Pain. Burns.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion		Do not eat, drink, or smoke during work.	

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Consult an expert! Personal protection: gas-tight chemical protection suit including self-contained breathing apparatus. Ventilation. Shut off cylinder if possible. Isolate the area until the gas has dispersed. NEVER direct water jet on liquid. Remove gas with fine water spray. Do NOT let this chemical enter the environment.	According to UN GHS Criteria  DANGER May cause or intensify fire; oxidizer Contains gas under pressure; may explode if heated Fatal if inhaled Causes severe skin burns and eye damage May cause respiratory irritation May cause damage to lungs through prolonged or repeated exposure if inhaled Very toxic to aquatic life Transportation UN Classification UN Hazard Class: 2.3; UN Subsidiary Risks: 5.1 and 8
STORAGE	
Fireproof if in building. Provision to contain effluent from fire extinguishing. Separated from food and feedstuffs. See Chemical Dangers. Cool. Dry. Keep in a well-ventilated room. Store in an area without drain or sewer access.	
PACKAGING	
Special insulated cylinder. Marine pollutant.	



Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.
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CHLORINE**ICSC: 0126****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

GREENISH-YELLOW COMPRESSED LIQUEFIED GAS WITH PUNGENT ODOUR.

Physical dangers

The gas is heavier than air.

Chemical dangers

The solution in water is a strong acid. It reacts violently with bases and is corrosive. The substance is a strong oxidant. Reacts violently with combustible substances and reducing agents. The substance reacts with most organic and inorganic compounds, causing fire and explosion hazard. Attacks metals, some forms of plastic, rubber and coatings.

Formula: Cl₂

Molecular mass: 70.9

Boiling point: -34°C

Melting point: -101°C

Solubility in water, g/100ml at 20°C: 0.7

Vapour pressure, kPa at 20°C: 673

Relative vapour density (air = 1): 2.5

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

Serious by all routes of exposure.

Effects of short-term exposure

Lachrymation. The substance is corrosive to the eyes, skin and respiratory tract. Rapid evaporation of the liquid may cause frostbite. Inhalation may cause asthma-like reactions. Inhalation may cause pneumonitis. Inhalation may cause lung oedema, but only after initial corrosive effects on eyes and/or airways have become manifest. See Notes. Exposure could cause death.

Inhalation risk

On loss of containment, a harmful concentration of this gas in the air will be reached very quickly.

Effects of long-term or repeated exposure

The substance may have effects on the respiratory tract and lungs. This may result in chronic inflammation and impaired functions. The substance may have effects on the teeth. This may result in erosion.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 0.1 ppm as TWA; 0.4 ppm as STEL; A4 (not classifiable as a human carcinogen).

MAK: peak limitation category: I(1); pregnancy risk group: C.

EU-OEL: 1.5 mg/m³, 0.5 ppm as STEL

ENVIRONMENT

The substance is very toxic to aquatic organisms. It is strongly advised not to let the chemical enter into the environment.

NOTES

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.

Immediate administration of an appropriate inhalation therapy by a doctor, or by an authorized person, should be considered.

The odour warning when the exposure limit value is exceeded is insufficient.

Do NOT use in the vicinity of a fire or a hot surface, or during welding.

Do NOT spray water on a leaking cylinder (to prevent corrosion of the cylinder).

Turn leaking cylinder with the leak up to prevent escape of gas in liquid state.

ADDITIONAL INFORMATION**EC Classification**

Symbol: T, N; R: 23-36/37/38-50; S: (1/2)-9-45-61

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